

RDMSWTN Mathematical Derivation and Implementation Notes

EMC2

May 3, 2026

Abstract

This document records the mathematics implemented by the EMC2 RDMSWTN code. The evidence process starts at non-decision time t_0 ; Erlang guess and kill clocks start at absolute trial time zero. The standard RDMSWTN path is a shifted Wald accumulator with optional uniform start-point variability A and truncated-normal drift variability sv . The current implementation uses closed forms for the canonical Wald, fixed-threshold SWTN, Wald start-point variability, the joint $A > 0, sv > 0$ density without Erlang clocks, exponential local guess–kill CDFs, and Erlang-2 local guess–kill CDFs when drift variability is absent. Numerical quadrature remains only where the current closed-form identities do not apply.

Contents

1	Parameterization and Time Domains	3
2	Fixed-Threshold SWTN	3
3	Uniform Start-Point Variability with Fixed Drift	4
4	Joint Start and Drift Variability Density	4
5	GBM Path	5
6	Erlang Guess and Kill Clocks	6
6.1	Kill-Only Evidence Density	6
6.2	Wald Exponential Tilting	6
7	Local Guess–Kill	7
7.1	Exponential Clocks	7
7.2	Erlang-2 Clocks without Drift Variability	7
8	CDFs, Omissions, and Race Likelihood Use	8

9	Current Closed-Form and Quadrature Boundary	8
10	Numerical Stability	8

1 Parameterization and Time Domains

Let T denote raw response time and let

$$x = T - t_0$$

be evidence-accumulation time. The evidence accumulator has zero density for $x \leq 0$. Erlang clocks, when present, are functions of raw time T , not evidence time x .

For the Wald accumulator, a fixed distance $d > 0$ is crossed by

$$X_t = \mu t + \sigma W_t.$$

The first-passage density is

$$f_W(x; d, \mu, \sigma) = \frac{d}{\sqrt{2\pi\sigma^2x^3}} \exp\left[-\frac{(d - \mu x)^2}{2\sigma^2x}\right], \quad x > 0. \quad (1)$$

The CDF is

$$F_W(x; d, \mu, \sigma) = \Phi\left(\frac{\mu x - d}{\sigma\sqrt{x}}\right) + \exp\left(\frac{2\mu d}{\sigma^2}\right) \Phi\left(-\frac{d + \mu x}{\sigma\sqrt{x}}\right). \quad (2)$$

The RDMSWTN model uses a threshold-distance representation. With uniform start-point variability,

$$d \sim U[b - A, b].$$

In EMC2 model construction, the user-facing parameter B is transformed to $b = B + A$, so the distance interval is $[B, B + A]$ after transformation. In the C++ density/CDF kernels the arguments are already the physical b and A .

Between-trial drift variability is

$$\xi \sim \mathcal{N}(\mu_0, sv^2) \text{ truncated to } \xi \geq c, \quad Z = P(\xi \geq c) = \Phi\left(\frac{\mu_0 - c}{sv}\right).$$

The default lower truncation in the R wrappers is $c = 0$ unless a column named `c` is supplied.

2 Fixed-Threshold SWTN

For fixed distance d and $sv > 0$, the SWTN density is

$$f_{\text{SWTN}}(x; d) = \frac{d}{\sqrt{2\pi x^3(sv^2x + \sigma^2)}} \exp\left[-\frac{(d - \mu_0 x)^2}{2x(sv^2x + \sigma^2)}\right] \frac{\Phi\left(\frac{\mu_* - c}{s_*}\right)}{Z}, \quad (3)$$

where

$$\mu_* = \frac{d sv^2 + \mu_0 \sigma^2}{sv^2x + \sigma^2}, \quad s_* = \sqrt{\frac{\sigma^2 sv^2}{sv^2x + \sigma^2}}. \quad (4)$$

This is implemented by `dswtn_core`. It follows from multiplying the Wald kernel by the normal drift density and completing the square in drift.

The no-clock fixed-threshold SWTN CDF is also closed form. It is represented as a sum of two bivariate-normal terms corresponding to the direct and image terms in the Wald CDF. The code selects the fast bivariate-normal approximation in ordinary regions and the more accurate branch in extreme tails or near-singular correlation.

3 Uniform Start-Point Variability with Fixed Drift

For $sv = 0$ and $A > 0$, the density and CDF are computed in the Wald path. The CDF is

$$F_A(x) = \frac{1}{A} \int_{b-A}^b F_W(x; d, \mu, \sigma) dd. \quad (5)$$

The implementation evaluates the canonical Wald-style primitive, including the special handling needed when the drift or tilt parameter is near zero. This path avoids PIGT-style parameterization in the public math and follows the ordinary Wald distance/drift notation.

4 Joint Start and Drift Variability Density

The standard no-clock RDMSWTN density with both uniform start-point variability and truncated-normal drift variability has a closed form. This replaces the older one-dimensional GL average of `dswtn_core` over thresholds.

Using

$$f_W(x; d, \mu, \sigma) = \frac{d}{x} \varphi(d; \mu x, \sigma^2 x), \quad (6)$$

where $\varphi(\cdot; m, s^2)$ is a normal density, define the latent Gaussian pair

$$\mu \sim \mathcal{N}(\mu_0, sv^2), \quad D_x = \mu x + \sigma \sqrt{x} \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1). \quad (7)$$

Then

$$E[D_x] = m_D = \mu_0 x, \quad \text{Var}(D_x) = s_D^2 = sv^2 x^2 + \sigma^2 x, \quad (8)$$

with correlation

$$\rho = \frac{sv\sqrt{x}}{\sqrt{\sigma^2 + sv^2 x}}. \quad (9)$$

Let

$$Y = \frac{D_x - m_D}{s_D}, \quad X = \frac{\mu - \mu_0}{sv}, \quad (10)$$

so (Y, X) is standard bivariate normal with correlation ρ . Define

$$\alpha_L = \frac{b - A - m_D}{s_D}, \quad \alpha_H = \frac{b - m_D}{s_D}, \quad \gamma = \frac{c - \mu_0}{sv}. \quad (11)$$

The density is the first moment of this bivariate normal over a rectangle:

$$f_{A,sv}(x) = \frac{1}{AZx} E[D_x \mathbf{1}\{\alpha_L \leq Y \leq \alpha_H, X \geq \gamma\}]. \quad (12)$$

Equivalently,

$$f_{A,sv}(x) = \frac{m_D P_R + s_D M_R}{AZx}, \quad (13)$$

where

$$P_R = [\Phi(\alpha_H) - \Phi(\alpha_L)] - [\Phi_2(\alpha_H, \gamma; \rho) - \Phi_2(\alpha_L, \gamma; \rho)] \quad (14)$$

and

$$M_R = \left[-\phi(y) \Phi\left(\frac{\rho y - \gamma}{\sqrt{1 - \rho^2}}\right) + \rho \phi(\gamma) \Phi\left(\frac{y - \rho \gamma}{\sqrt{1 - \rho^2}}\right) \right]_{\alpha_L}^{\alpha_H}. \quad (15)$$

The coefficient in the second endpoint term is $\rho \phi(\gamma)$; the apparent extra factor of $(1 - \rho^2)^{-1/2}$ cancels when integrating the product of the two Gaussian densities.

This identity is implemented by `drdmswt_n_joint_A_sv_density` and is used only when $\lambda_g = \lambda_k = 0$ and the call is a density, not a CDF. In a benchmark of 5,000 no-clock joint $A + sv$ density evaluations, the old 20-node threshold GL construction took 2.722 seconds, while the closed form took 0.370 seconds, a 7.36-fold speedup. The maximum absolute difference from the GL20 reference was 9.55×10^{-7} and the median relative difference was 1.51×10^{-10} .

5 GBM Path

For the GBM accumulator

$$dY_t = \mu Y_t dt + \sigma Y_t dW_t, \quad Y_0 = y_0, \quad (16)$$

let $Z_t = \log Y_t$. Then

$$dZ_t = \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t. \quad (17)$$

Thus GBM first passage is an ordinary Wald first-passage problem in log space with drift

$$\nu = \mu - \frac{\sigma^2}{2} \quad (18)$$

and distance

$$a = \log\left(\frac{b}{y_0}\right). \quad (19)$$

The important implementation detail is that A is physical-space start variability, not log-space half-width. If $Y_0 \sim U[1, 1 + A]$, then the corresponding log-distance integral has the Jacobian e^y/A . The code accounts for this by transforming the physical bounds into log bounds and applying the Jacobian-weighted primitive. The same tilted-Wald identities used for Wald apply conditionally on the log-distance and log-drift.

6 Erlang Guess and Kill Clocks

For an Erlang clock with shape n and rate λ ,

$$f_K(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{(n-1)!}, \quad S_K(t) = e^{-\lambda t} \sum_{j=0}^{n-1} \frac{(\lambda t)^j}{j!}. \quad (20)$$

Only $n = 1$ and $n = 2$ are special-cased analytically in the current model code.

6.1 Kill-Only Evidence Density

If the clock kills but does not generate an observed response, the defective evidence density is

$$g_n(T) = \mathbf{1}\{T > t_0\} f_X(T - t_0) S_K(T). \quad (21)$$

For an exponential clock,

$$g_1(T) = \mathbf{1}\{T > t_0\} e^{-\lambda t_0} f_X(x) e^{-\lambda x}, \quad x = T - t_0. \quad (22)$$

For Erlang-2,

$$g_2(T) = \mathbf{1}\{T > t_0\} e^{-\lambda(t_0+x)} f_X(x) [1 + \lambda(t_0 + x)]. \quad (23)$$

6.2 Wald Exponential Tilting

For fixed distance and drift, define

$$q_r = \sqrt{\mu^2 + 2\sigma^2 r}, \quad L(r) = \exp\left[\frac{d}{\sigma^2}(\mu - q_r)\right]. \quad (24)$$

Then

$$f_W(x; d, \mu, \sigma) e^{-rx} = L(r) f_W(x; d, q_r, \sigma). \quad (25)$$

The zeroth tilted moment over a finite window is

$$M_0(U; r) = \int_0^U f_W(x; d, \mu, \sigma) e^{-rx} dx = L(r) F_W(U; d, q_r, \sigma). \quad (26)$$

The first moment satisfies

$$M_1(U; r) = \int_0^U x f_W(x; d, \mu, \sigma) e^{-rx} dx = -\frac{\partial}{\partial r} M_0(U; r). \quad (27)$$

The finite-window second moment used by the Erlang-2 local guess-kill path is

$$M_2(U; r) = \frac{d^2}{q_r^2} M_0(U; r) + \frac{\sigma^2}{q_r^2} M_1(U; r) - L(r) \frac{2d\sigma\sqrt{U}}{q_r^2} \phi(d_-), \quad (28)$$

where

$$d_- = \frac{q_r U - d}{\sigma\sqrt{U}}. \quad (29)$$

As $U \rightarrow \infty$,

$$M_0(\infty; r) = L(r), \quad M_1(\infty; r) = \frac{d}{q_r} L(r), \quad M_2(\infty; r) = L(r) \left(\frac{d^2}{q_r^2} + \frac{d\sigma^2}{q_r^3} \right). \quad (30)$$

7 Local Guess–Kill

Let G be the local guess clock and K the local kill clock. The observed response density for one accumulator is

$$p(T) = f_D(T)S_G(T)S_K(T) + f_G(T)S_K(T)S_D(T), \quad (31)$$

where $D = t_0 + X$ and $S_D(T) = P(D > T)$.

7.1 Exponential Clocks

For exponential clocks with rates λ_g and λ_k , let

$$r = \lambda_g + \lambda_k. \quad (32)$$

The local guess–kill CDF is

$$P(Y \leq T) = D_r(T) + \lambda_g \frac{1 - e^{-rT} S_D(T) - D_r(T)}{r}, \quad (33)$$

where

$$D_r(T) = \int_0^T f_D(t) e^{-rt} dt. \quad (34)$$

In the code this is `local_combo_response_cdf_exp`. The term $D_r(T)$ is evaluated by the killed-Wald or killed-GBM CDF at the combined rate r .

7.2 Erlang-2 Clocks without Drift Variability

For Erlang-2 local guess–kill with rates λ_g and λ_k ,

$$S_G(T)S_K(T) = e^{-rT}(1 + \lambda_g T)(1 + \lambda_k T) = e^{-rT} [1 + rT + \lambda_g \lambda_k T^2]. \quad (35)$$

Writing $T = t_0 + x$ gives the decision contribution

$$e^{-rt_0} [a_0 M_0(U; r) + a_1 M_1(U; r) + a_2 M_2(U; r)], \quad (36)$$

where $U = T - t_0$ and

$$a_0 = 1 + rt_0 + \lambda_g \lambda_k t_0^2, \quad a_1 = r + 2\lambda_g \lambda_k t_0, \quad a_2 = \lambda_g \lambda_k. \quad (37)$$

The guess contribution is split into the raw-time interval $[0, t_0]$, where $S_D(T) = 1$, and the post- t_0 interval. The pre- t_0 piece is an elementary polynomial-exponential integral. The post- t_0 piece is reduced by integration by parts to the same elementary integrals and M_0, M_1, M_2 .

This closed form is implemented for fixed-drift Wald, ordinary Wald SPV, and GBM physical-space SPV. It is not used when $sv > 0$, because drift mixing introduces nonlinear $q(\mu) = \sqrt{\mu^2 + 2\sigma^2 r}$ terms that do not remain in the bivariate-normal family.

8 CDFs, Omissions, and Race Likelihood Use

For a kill-only model, an observed evidence response at raw time T contributes

$$\log g_n(T), \quad (38)$$

and an omission by cutoff C contributes

$$\log [1 - G_n(C)]. \quad (39)$$

For response-generating guess clocks, the observed density uses the sum of evidence and guess densities, and no-response probability by cutoff is the joint survivor of the decision and clock races.

In multi-accumulator race likelihoods, each accumulator supplies a density and CDF/survivor at the trial time. Logical-rule models may map several latent accumulator outcomes to one observed response; those mappings are handled in the particle likelihood layer and do not change the single-accumulator formulas in this document.

9 Current Closed-Form and Quadrature Boundary

Case	Density	CDF
Fixed Wald, no clocks	closed form	closed form
Wald with $A > 0$, $sv = 0$	closed form	closed form
Fixed-threshold SWTN, $A = 0$, $sv > 0$, no clocks	closed form	closed form
Joint $A > 0$, $sv > 0$, no clocks	closed form	GL over drift for CDF
Kill-only exponential/Erlang-2, $sv = 0$	closed form	closed form
Kill-only with $sv > 0$	closed-form density, GL CDF	GL over drift
Local guess-kill exponential	closed-form density	closed form
Local guess-kill Erlang-2, $sv = 0$	closed-form density	closed form
Local guess-kill Erlang-2, $sv > 0$	raw-time GL bridge	raw-time GL bridge
GBM physical SPV, no drift variability	closed form	closed form where implemented

The guiding rule is simple: the implementation uses analytic Wald, tilted-Wald, GBM log-transform, and bivariate-normal identities wherever the transformed integrals remain in those families. It uses GL quadrature only for the remaining one-dimensional mixtures where the transformed integrand contains the Wald image term with a bilinear drift-distance interaction, or Erlang tilting with the nonlinear $q(\mu)$ drift transform.

10 Numerical Stability

The implementation uses stable normal tail calls, bivariate-normal branch selection, signed-log arithmetic for primitive differences, and explicit small-parameter branches where cancellation is otherwise severe. The Erlang and tilted-moment paths avoid direct subtraction

of nearly equal large terms where possible, and clamp final probabilities into their valid ranges only after the mathematically meaningful computation is complete.